ITT300 · GROUP PROJECT · 2026

Enterprise Network Design

A Cisco Packet Tracer project on LAN topology design, IP addressing, static routing, DHCP and DNS/HTTP services

Prepared by

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Diploma in Computer Science · UiTM Jasin

Course: ITT300 — Introduction to Data Communication and Networking

This was completed as a group project by a team of four. This portfolio presents the completed network build and my understanding of the work.

1. PROJECT OVERVIEW

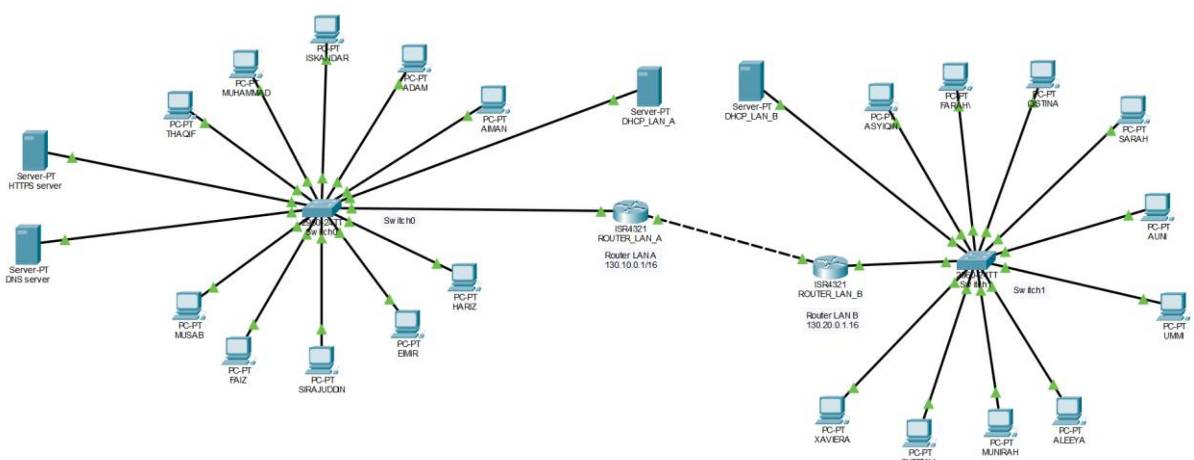
This portfolio documents a group project completed for ITT300: Introduction to Data Communication and Networking, in which Cisco Packet Tracer was used to design, configure, and test a small enterprise network. The objective was to apply core networking concepts covered in the course, including network topology design, IP addressing, static routing between routers, DHCP-based address allocation, and DNS/HTTP service configuration, within a simulated environment.

The network consists of two Local Area Networks, LAN A and LAN B, each built around a switch connecting ten PCs and linked to the wider network through a router. The two routers are connected to each other to enable inter-subnet communication. The sections below walk through the network design, the LAN topologies, router configuration, DHCP setup, and DNS/HTTP services, with Packet Tracer screenshots as evidence of successful configuration and testing at each stage.

Platform	Cisco Packet Tracer (network simulation)
Network scale	2 Routers, 2 Switches, 20 PCs (10 per LAN)
Services configured	Static routing, DHCP, DNS, HTTP
Project type	Group project — team of 4 students
Course	ITT300 — Introduction to Data Communication and Networking

2. NETWORK DESIGN

One network design with 10 personal computers connected to each switch, 2 switches, and 2 routers.

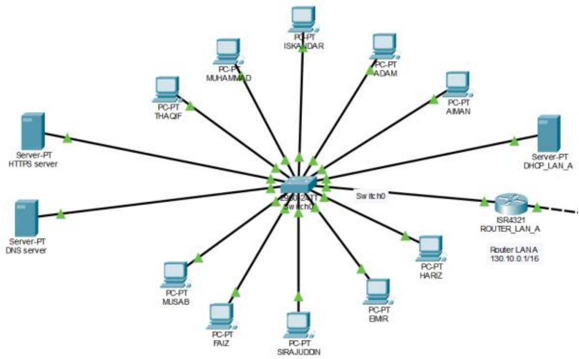


Completed network topology

Two routers, two switches, and ten PCs connected to each switch form the overall network design.

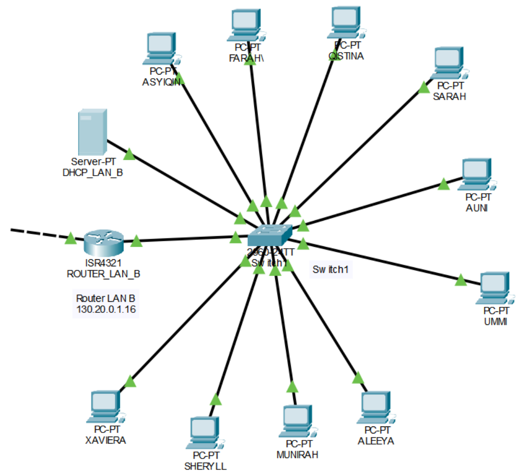
3. LAN TOPOLOGY & CONNECTIVITY

Two identical LAN structures — LAN A and LAN B — were built and verified for correct addressing and inter-LAN connectivity.



LAN A topology

PCs, switch, and router interconnected within LAN A.



LAN B topology

The equivalent network arrangement built for LAN B.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 130.20.0.100/16
Ping request could not find host 130.20.0.100/16. Please check the name and try again.
C:\>ping 130.20.0.100

Pinging 130.20.0.100 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 130.20.0.100: bytes=32 time<1ms TTL=126
Reply from 130.20.0.100: bytes=32 time<1ms TTL=126

Ping statistics for 130.20.0.100:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Successful ping: LAN A → LAN B

Confirms that devices in LAN A can communicate with devices in LAN B.

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::20D:BDFF:FE12:DEA4
   IPv6 Address....................: ::
   IPv4 Address....................: 130.10.0.104
   Subnet Mask.....................: 255.255.0.0
   Default Gateway.................: ::
                                     130.10.0.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

LAN A PC ipconfig

Verifies the IP address, subnet mask, and default gateway assigned to a PC in LAN A.

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::204:9AFF:FE7D:178E
   IPv6 Address....................: ::
   IPv4 Address....................: 130.20.0.100
   Subnet Mask.....................: 255.255.0.0
   Default Gateway.................: ::
                                     130.20.0.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

LAN B PC ipconfig

Verifies the IP configuration assigned to a PC in LAN B.

4. ROUTER CONFIGURATION

Both routers were assigned the correct interface IP addresses and linked together to allow inter-subnet routing.

```
Router>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   130.10.0.1      YES manual up                    up
GigabitEthernet0/0/1   130.30.0.1      YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

Router 0 IP configuration

Confirms Router 0 has been assigned the correct IP address on its interface.

```
Router>show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   130.20.0.1      YES manual up                    up
GigabitEthernet0/0/1   130.30.0.2      YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

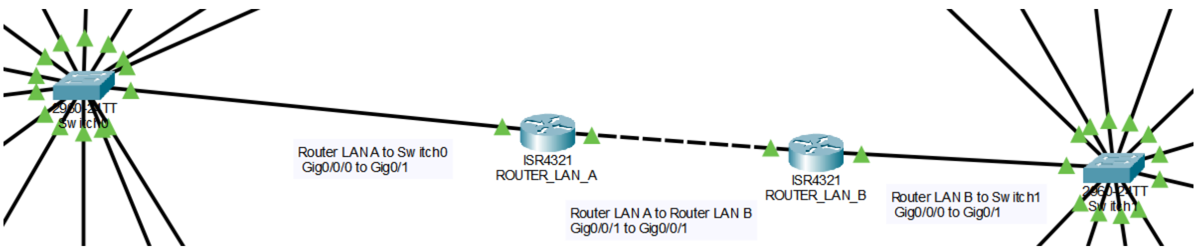
Router 1 IP configuration

Confirms Router 1 has been assigned the correct IP address on its interface.

```
Router>ping 130.10.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 130.10.0.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
```

Ping: Router 1 → Router 0

Verifies that Router 1 can successfully reach Router 0.



Routers connected

The two routers connected to each other, forming the link between LAN A and LAN B.

5. DHCP CONFIGURATION

DHCP pools were configured on both LANs so that PCs automatically receive IP addresses instead of manual assignment.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask
serverPool	130.10.0.1	130.10.0.21	130.10.0.100	255.255.0.0

DHCP pool — LAN A

Sets up the DHCP pool that automatically assigns IP addresses to devices in LAN A.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask
serverPool	130.20.0.1	130.10.0.21	130.20.0.100	255.255.0.0

DHCP pool — LAN B

Sets up the DHCP pool that automatically assigns IP addresses to devices in LAN B.

```
FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::20D:BDFF:FE12:DEA4
   IPv6 Address....................: ::
   IPv4 Address....................: 130.10.0.104
   Subnet Mask.....................: 255.255.0.0
   Default Gateway.................: ::
                                     130.10.0.1
```

LAN A auto-assigned IP

Confirms a PC in LAN A obtained an IP address automatically from the DHCP pool.

```
FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::204:9AFF:FE7D:178E
   IPv6 Address....................: ::
   IPv4 Address....................: 130.20.0.100
   Subnet Mask.....................: 255.255.0.0
   Default Gateway.................: ::
                                     130.20.0.1
```

LAN B auto-assigned IP

Confirms a PC in LAN B obtained an IP address automatically from the DHCP pool.

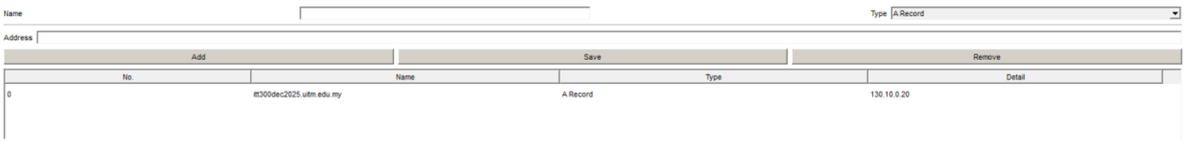
```
C:\>ping 130.20.0.100

Pinging 130.20.0.100 with 32 bytes of data:

Reply from 130.20.0.100: bytes=32 time<1ms TTL=126
Reply from 130.20.0.100: bytes=32 time<1ms TTL=126
Reply from 130.20.0.100: bytes=32 time<1ms TTL=126
Reply from 130.20.0.100: bytes=32 time<1ms TTL=126

Ping statistics for 130.20.0.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Inter-subnet ping proof

Proves that devices on different subnets can communicate with one another through the routers.

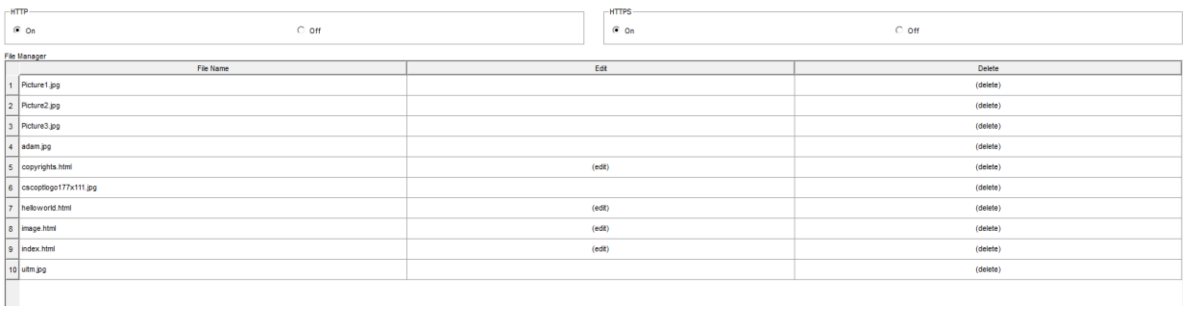
6. DNS & HTTP SERVICES

A DNS record and an HTTP server were configured so the network's web page could be reached either by domain name or by IP address.



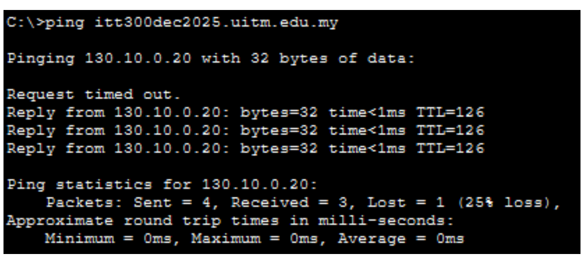
DNS server configuration

The DNS server configuration used to resolve the domain name to its IP address.



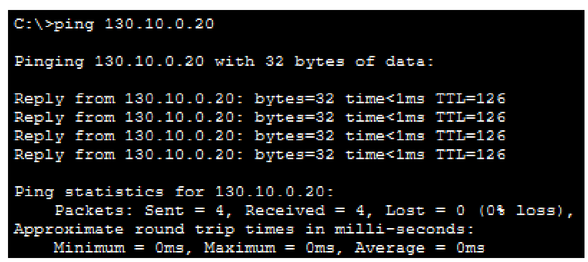
HTTP server edit

The HTTP server settings being edited to host the web page.



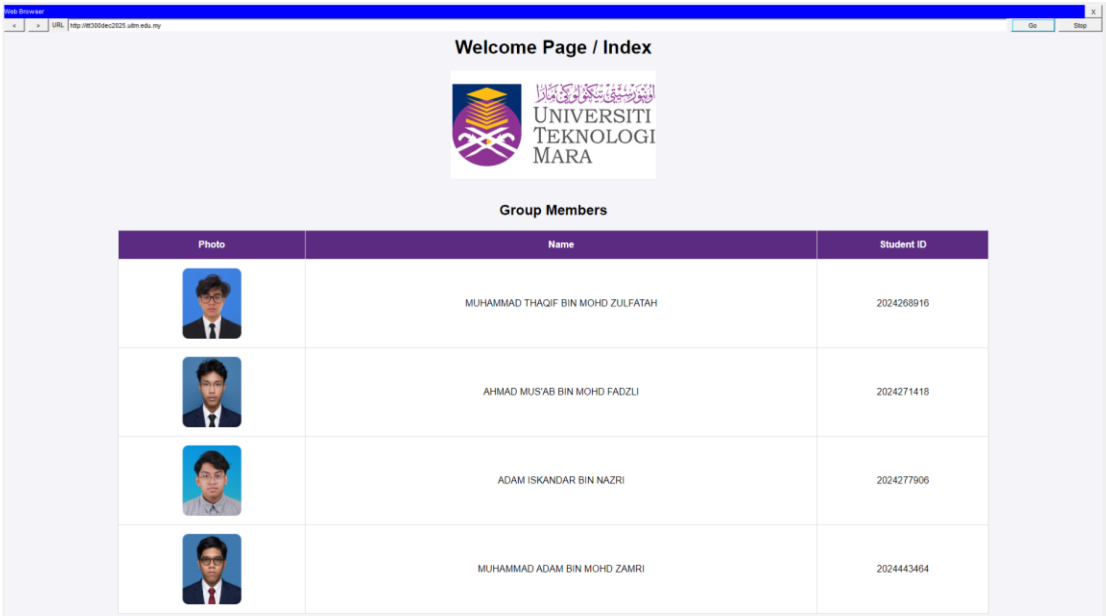
Ping by domain name

Confirms the domain name successfully resolves to its corresponding IP address.



Ping by IP address

Confirms connectivity to the server using its IP address directly.



Web page — index.html

The web page as displayed in the browser.

File Name: index.html

```html
<html>
<head>
  <title>Welcome Page / Index</title>
  <style>
    body { font-family: Arial, sans-serif; text-align: center; background-color: #f4f4f9; }
    .logo { width: 350px; margin-bottom: 20px; }
    .student-photo { width: 100px; height: 120px; border-radius: 10px; }
    table { margin: 0 auto; background-color: white; border-collapse: collapse; width: 80%; }
    th, td { border: 1px solid #ddd; padding: 15px; text-align: center; font-size: 16px; }
    th { background-color: #5b2b82; color: white; }
  </style>
</head>
<body>

  <h1>Welcome Page / Index</h1>

  <img src="uitm.jpg" alt="UiTM Logo" class="logo">

  <h2>Group Members</h2>
  <table>
    <tr>
      <th>Photo</th>
      <th>Name</th>
      <th>Student ID</th>
    </tr>
    <tr>
      <td><img src="Picture1.jpg" alt="Thaqif" class="student-photo"></td>
      <td>MUHAMMAD THAQIF BIN MOHD ZULFATAH</td>
      <td>2024268916</td>
    </tr>
    <tr>
      <td><img src="Picture2.jpg" alt="Mus'ab" class="student-photo"></td>
      <td>AHMAD MUS'AB BIN MOHD FADZLI</td>
      <td>2024271418</td>
    </tr>
    <tr>
      <td><img src="Picture3.jpg" alt="Adam Iskandar" class="student-photo"></td>
      <td>ADAM ISKANDAR BIN NAZRI</td>
      <td>2024277906</td>
    </tr>
    <tr>
      <td><img src="adam.jpg" alt="Muhammad Adam" class="student-photo"></td>
      <td>MUHAMMAD ADAM BIN MOHD ZAMRI</td>
      <td>2024443464</td>
    </tr>
  </table>

</body>
</html>
```

index.html coding

The HTML code used to build the web page.

7. SKILLS DEMONSTRATED

- Network topology design — planning and building a two-LAN enterprise network in Cisco Packet Tracer
- IP addressing & subnetting — assigning and verifying host, gateway, and interface addresses
- Static routing — connecting two routers to enable inter-subnet communication
- DHCP configuration — setting up address pools for automatic IP assignment per LAN
- DNS configuration — mapping a domain name to a server IP address
- HTTP service hosting — configuring and editing a server to host a web page
- HTML — writing the markup for the hosted index.html page
- Network troubleshooting — using ping and ipconfig to verify connectivity and configuration
- Team collaboration — dividing tasks and integrating individual work into one shared network build

8. ABOUT ME

Full name	Adam Iskandar bin Nazri
Student ID	2024277906
Programme	Diploma in Computer Science
Institution	UiTM Jasin
Course	ITT300 — Introduction to Data Communication and Networking
Project type	Group project (team of 4 students)
Year	2026

This project was completed collaboratively as part of a four-member team. This portfolio reflects the completed network build and my understanding of the design, configuration, and testing process carried out for the assignment.